

TB-LiteNet tuberculosis detection with a lightweight scratch-trained CNN using heterogeneous chest X-ray data

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ABSTRACT

Background and Aim: Tuberculosis (TB), caused by *Mycobacterium tuberculosis*, continues to present a major health challenge, particularly in low- and middle-income countries with constrained diagnostic resources. Although chest radiography remains a standard screening method, the shortage of trained radiologists and variability in image interpretation hinder early and reliable detection. This study aimed to develop a lightweight convolutional neural network (CNN), trained from scratch with Swish activation, for accurate and efficient TB detection from chest radiographs.

Patients and Methods: A heterogeneous dataset was created by aggregating multiple publicly available repositories, including Montgomery, Shenzhen, TB Base, and TBX11K. Images were standardized through grayscale conversion, resizing, and normalization, and subsequently partitioned into stratified training, validation, and test sets. The proposed CNN consisted of multiple convolutional layers with Swish activation and max pooling, followed by dense layers with dropout regularization. Model optimization was performed using the Adam optimizer, and performance was evaluated using accuracy, precision, recall, and F1-score metrics. External validation was conducted using unseen and independent datasets to assess generalizability.

Results: The model achieved an overall accuracy of 92%, with precision and recall values exceeding 0.90 and an F1-score of 0.92. External validation confirmed the model's generalizability, with inference times averaging 30–40 ms (specifically utilizing the Intel i5 CPU for inference) per image on consumer-grade hardware, demonstrating computational efficiency.

Conclusions: The findings demonstrate that a lightweight, scratch-trained CNN can provide accurate and computationally efficient TB detection from chest radiographs. The deliberate choice of dataset heterogeneity and lightweight architecture allows strong diagnostic performance without reliance on high-end hardware, making the system suitable for TB screening programs in resource-limited healthcare environments.

KEYWORDS

tuberculosis detection, chest X-ray analysis, lightweight convolutional neural network, deep learning in medical imaging, low-resource healthcare deployment

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1. Introduction

Tuberculosis (TB) is caused by *Mycobacterium tuberculosis*, and it remains one of the most persistent global health threats. It is both preventable and curable, however TB continues to cause high mortality worldwide, particularly in middle to low-income countries where healthcare infrastructure is constrained. Pakistan ranks among the five countries with the highest tuberculosis (TB) burden worldwide and contributes approximately 6.3% to the global number of TB cases [1]. The estimated TB incidence in Pakistan varies significantly across districts from 110 to 462 per 100,000 population per year, with Sindh at 292 (95% CI:

213–384), Khyber Pakhtunkhwa at 269 (95% CI: 196–354), and Punjab at 243 (95% CI: 177–320) per 100,000 per year [2]. In 2022, around 10.6 million individuals developed active tuberculosis, with nearly 1.3 million deaths reported worldwide, positioning TB as the second leading cause of death from a single infectious agent after COVID-19 [3]. The challenge lies not only in treatment but also in early detection, as delayed or missed diagnoses fuel ongoing transmission and increase mortality rates. Chest radiography continues to serve as one of the most accessible and cost-effective methods for pulmonary TB screening. However, interpretation requires trained radiologists, whose availability is limited in many high-burden regions. Moreover, radiographic interpretation is subject to inter-reader variability, and large-scale screening programs are often constrained by time and personnel requirements [4, 5].

In response, artificial intelligence (AI) has been increasingly integrated into medical imaging, with CNNs at the forefront of automated TB detection. CNN-based models excel at learning hierarchical image features from raw pixel data, eliminating the need for handcrafted feature engineering. Several studies demonstrate that CNNs can achieve competitive diagnostic performance and sometimes surpassing expert radiologists in TB classification. For instance, deep learning models applied to chest radiographs have shown remarkable accuracy and reproducibility, providing a scalable alternative to manual interpretation [6–9].

Despite these advancements, challenges remain. Many high-performing models require extensive computational resources, including high-end GPUs, which limit their deployment in constrained healthcare environments. Furthermore, much existing research relies on homogeneous datasets, often sourced from a single repository. While these datasets enable proof-of-concept validation, they may not adequately reflect the heterogeneity of clinical imaging, which varies by acquisition device, imaging quality, and annotation standards. Thus, the models that are trained on such datasets often struggle to generalize to new clinical contexts.

Additionally, while attention-based and transformer-driven models achieve impressive results, their complexity raises questions of reproducibility, interpretability, and real-world applicability in high-burden but resource-limited settings [4, 5]. Complementary work has also shown that even ensembles of scratch-trained CNNs can deliver strong performance, provide sound results without resorting to computationally expensive transformer backbones [10].

The contributions of this study are as follows:

- A lightweight Convolutional Neural Network (CNN) was developed and trained from scratch, focusing on maintaining diagnostic accuracy while keeping computational requirements modest.
- A heterogeneous dataset was assembled by integrating multiple publicly available repositories, allowing the model to learn from varied imaging conditions and annotation standards.

- The model was assessed through external validation and inference on consumer-grade hardware to evaluate generalizability and feasibility for use in resource-constrained settings.

2. Patients & methods

This study used publicly available anonymized datasets and did not require institutional ethical approval. The study complies with the Declaration of Helsinki.

To mitigate dataset bias and enhance the generalization of extracted features, the data pool in this study was constructed by merging multiple publicly available TB CXR datasets rather than relying on a single source. This heterogeneity introduces variations in acquisition protocols, imaging quality, and annotation styles, which provide the model with more diverse feature distributions during training.

Prior research has shown that CNNs trained on multi-source datasets achieve stronger generalization to unseen clinical data, as heterogeneous inputs help the network to learn invariant, pathology-relevant features not just dataset-specific artifacts [11]. Complementary findings by Irhamsyah et al. [12] highlight that combining multiple TB CXR datasets such as Montgomery, Shenzhen, and Belarus significantly impacted the performance of deep learning models, further justifying multi-dataset approach adopted here.

- Montgomery County Dataset (USA): This dataset, released by the U.S. National Library of Medicine (NLM), consists of 138 chest X-ray images, including 80 TB-negative and 58 cases with TB manifestations, accompanied by manual lung segmentation masks and clinical annotations [13].
- Shenzhen Dataset (China): The Shenzhen Hospital TB dataset contains 662 chest X-ray images, of which 326 are TB-negative and 336 exhibit TB-related abnormalities. Clinical annotations are also included for reference [14].
- TB Chest Radiography Database: To supplement the above datasets, the TB Chest Radiography Database introduced by Rahman et al. [15] was also referenced. This collection has been widely used in the literature for benchmarking deep learning-based TB detection.
- TBX11K Dataset: A large dataset comprising 11,200 X-ray images, TBX11K includes bounding box annotations across five categories: healthy, sick non-TB, active TB, latent TB, and uncertain. It provides extensive coverage for model generalization and evaluation [16].

The datasets for external validation consisted of the following:

- Tuberculosis (TB) Chest X-ray Database (Roboflow): This open-source dataset, hosted on the Roboflow Universe platform by COVID19 XRAYs, contains chest radiograph images curated for TB detection research, including 37 normal and 37 TB-positive. The collection includes labeled TB-positive and TB-negative cases, providing a diverse set of radiographs suitable for model

training and evaluation. The dataset is public and can be used in computer vision projects for medical imaging applications [17].

- **Mendeley Tuberculosis Chest X-rays Dataset:** This dataset, published on Mendeley Data by Kiran and Jabeen (2024), provides a collection of chest radiograph images specifically curated for TB detection research. The dataset includes 2,494 TB-positive and 514 TB-negative cases. Its structured organization and open accessibility make it a valuable resource for benchmarking algorithms in medical imaging [18].

In this study, we utilized a grand total of 2,114 chest radiographs. We constructed a unified internal dataset by cleaning and combining 1,894 images from four separate public repositories (Table 1). The dataset was subsequently divided into stratified training (70%), internal testing (20%), and validation (10%) subsets. As detailed in Table 1, the raw testing partition contained 379 TB-positive images. However, to ensure a statistically balanced evaluation between classes, the final test set was standardized to **512 images**, comprising **256 TB-positive images** (randomly sampled from the testing pool) and **256 Normal images**. This balanced composition prevents class bias in the performance metrics presented in Table 2. This external set, comprising 20 diverse pilot images and 200 images from the Mendeley dataset, was kept separate and unseen during the model training process.

2.1. Data pre-processing

The images passed through a standardized pre-processing pipeline. Images were at first converted to grayscale to reduce computational complexity by eliminating redundant color channels, consistent with best practices in radiographic analysis [19]. The images were resized to 224 x 224 pixels that is standard input dimension in CNN architectures. Pixel intensity normalization was then applied using min-max scaling to [0,1], ensuring consistency across samples and stabilizing gradient descent [20].

Formula for normalization:

$$\frac{x - x_{min}}{x_{max} - x_{min}} \quad (1)$$

Finally, the dataset was divided into training, validation, and testing subsets using stratified sampling to preserve class balance.

Table 1. Composition of the heterogeneous training dataset

Dataset	TB count	% of TB pool	TB for train	TB for test	TB for Val
Montgomery	58	3.06%	41	11	6
Shenzhen	336	17.74%	235	67	34
TB Base	700	36.96%	518	132	50
TBX11k_2	785	42.24%	532	169	84
Total	1,879	100%	1,326	379	174

Composition of the heterogeneous dataset. *Note:* The 'TB for Test' column indicates the total available images (379). For the final performance evaluation (Table 2), a balanced subset of 256 TB-positive images was drawn from this pool and combined with 256 Normal images (Total $N = 512$).

Table 2. Model performance metrics on the internal test set

	Precision	Recall	F1	Support
Normal	0.91	0.94	0.92	256
TB	0.94	0.91	0.92	256

The table summarizes the precision, recall, F1-score, and support for both Normal and TB classes.

2.2. Model architecture

A custom CNN was designed for binary TB classification. The architecture was constructed to balance representational capacity with computational efficiency, following principles in medical imaging CNN design [21, 22]. The input layer ($224 \times 224 \times 1$) ensures reduced memory footprint without sacrificing diagnostic value [19]. The feature extractor consisted of three sequential convolutional blocks: 32, 64, and 128 filters with 3×3 kernels, each followed by Swish activation and max pooling. This progressive increase allows hierarchical feature extraction, from edges to complex pathological patterns [23].

The final block was flattened into a dense layer of 128 units, followed by dropout regularization (0.5) to minimize overfitting [24]. The output layer uses sigmoid activation for binary prediction. Swish activations were employed to improve gradient flow, outperforming ReLU in several biomedical contexts [25]. Adam optimizer with learning rate 1×10^{-4} was used [26], and binary cross-entropy loss guided optimization. Accuracy, precision, and recall were prioritized as evaluation metrics due to their statistical and clinical significance.

This design emphasizes lightweight deployment: inference can be executed in real time on mid-range CPUs (Intel i5, ≥ 8 GB RAM) without requiring high-end GPUs. This contrasts with complex architectures like TB-Net [7], which achieves higher raw accuracy but demand significantly greater computational resources. Thus, the proposed model favors reproducibility, accessibility, and feasibility in resource-constrained settings (Fig. 1).

2.3. Hyperparameter Configuration

- **Batch Size:** Set to **32** to balance computational efficiency and gradient updates.
- **Epochs:** The model was trained for **20** epochs, which allowed for convergence without overfitting, with early stopping employed to keep the optimal weights.

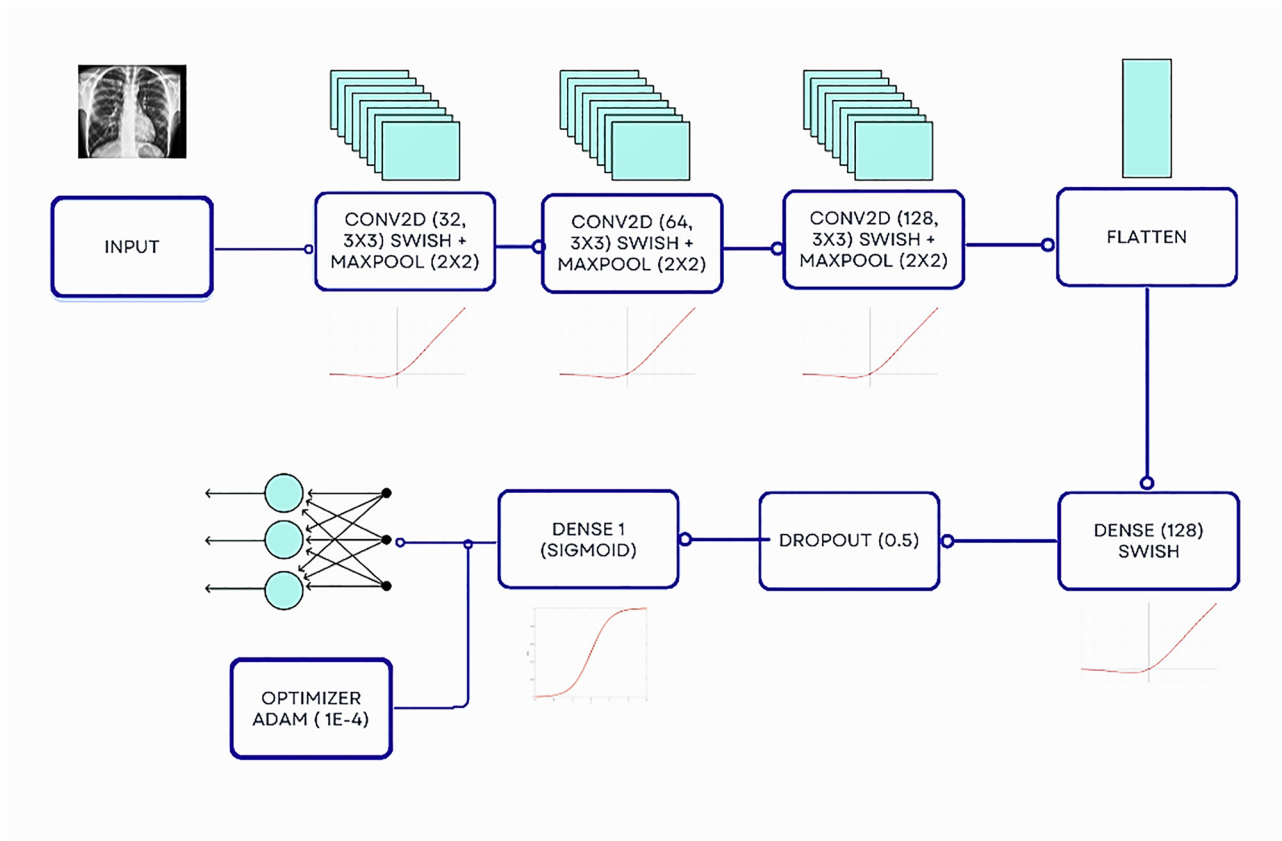


Fig. 1. The architecture of the proposed lightweight Convolutional Neural Network (CNN). The model takes a 224×224 grayscale chest X-ray as input. It consists of three successive convolutional blocks with increasing filter sizes (32, 64, 128), each with Swish activation and Max Pooling (2x2). The extracted feature maps are flattened and passed to the classifier head. This head is composed of a dense layer with 128 units and Swish activation, followed by a dropout layer (0.5 rate) for regularization. The final output layer uses the sigmoid activation function to produce a binary prediction

- **Optimizer:** The **Adam optimizer** was used with the following settings: $\text{lr} = 1 \times 10^{-4}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 1 \times 10^{-7}$, providing stable and adaptive gradient updates.
- **Loss Function:** A **binary cross-entropy** loss function was selected as suitable for binary probabilistic classification tasks.
- **Activations:** **Swish** was used for the hidden layers to ensure smooth gradient flow, while **Sigmoid** was used for the output layer to generate the final probability.
- **Regularization:** **Dropout** with a rate of **0.5** was applied to mitigate overfitting.
- **Initialization:** **Glorot/Xavier** initialization was used to keep gradients stable at the start of training.
- **Validation Split:** A **10%** validation split was used during each epoch to monitor the model's generalization performance throughout training.
- **Checkpointing:** The model weights with the **best validation accuracy** were saved to preserve the optimal model state.

2.4. Evaluation

Evaluation employed accuracy, precision, recall, specificity, and F1-score, supplemented by AUC-ROC. Confusion matrices visualized true/false classifications. In TB detection,

recall (sensitivity) carries clinical weight as false negatives are more harmful than false positives [27].

- Accuracy

$$\frac{(TP + TN)}{(TP + FN) + (FP + TN)} \quad (2)$$

- Precision

$$\frac{(TP)}{(TP + FP)} \quad (3)$$

- Specificity

$$\frac{(TN)}{(FP + TN)} \quad (4)$$

- F1

$$\frac{(2 * TP)}{(2 * TP + FN + FP)} \quad (5)$$

- Recall

$$\frac{(TP)}{(TP + FN)} \quad (6)$$

Where:

- **TP = True Positive**

- TN = True Negative
- FP = False Positive
- FN = False Negative

2.5. Implementation environment & workflow

Experiments were conducted on a workstation with Intel® Core™ i5 (6 cores, 12 threads), 16 GB RAM, and NVIDIA® GeForce® GTX 1660 Super GPU.

2.6. External validation

To establish an initial empirical performance baseline, the model's efficacy was first tested using a single-image analysis mode. At this stage, we selected a small but diverse set of 20 chest radiographs completely unseen during training. These were drawn from multiple sources, including the Mendeley Tuberculosis Chest X-ray dataset, samples from the Roboflow TB collection, publicly available Google-sourced images, and a subset of our own unseen validation data.

Each image was processed individually, with performance metrics such as prediction label, confidence score, and inference time systematically recorded. This step was intended to test the responsiveness and stability of the model under controlled conditions before moving toward large-scale evaluation. Following the preliminary assessment, the workflow was scaled up to incorporate the batch image processing system, enabling simultaneous evaluation of larger datasets. For this phase, we designed a rigorous test protocol using 200 radiographs, evenly split into 100 TB-positive and 100 TB-negative cases, all sourced from the Mendeley dataset. The batch evaluation allowed us to test both the classification pipeline and the automated export functionality.

The model's predicted labels associated confidence levels, and inference times for each image were programmatically exported, creating a reliable basis for downstream quantitative analysis. The outputs from both phases were aggregated and used to generate formal evaluation metrics.

A confusion matrix was utilized to quantify true positives, true negatives, false positives, and false negatives. Additionally, a Receiver Operating Characteristic (ROC) curve was constructed to further evaluate classification efficacy, providing a visual representation of the model's discriminative ability across decision thresholds and explicitly highlighting the trade-off between sensitivity and specificity. Together, these results form the foundation of the model's external validation, with all visualizations and numerical outcomes systematically presented in the Results section of this study.

3. Results

Proposed model TB-LiteNet was evaluated on the combined dataset using standard performance metrics across 20 epochs. Accuracy, precision, recall, and F1-score were computed to assess both statistical performance and clinical

relevance, with particular emphasis on recall given the risks associated with false negatives in tuberculosis screening [27].

The model achieved a precision of **0.91**, recall of **0.94**, and an F1-score of **0.92** for the Normal class, based on **256** samples. Similarly, for the TB class, the model attained a precision of **0.94**, recall of **0.91**, and an F1-score of **0.92**, also evaluated on **256** samples. These results indicate that the model demonstrates a balanced and consistent performance across both classes, with high precision and recall values (Fig. 2).

3.1. Results for external validation

Here are the results of the model's external evaluation. The first set corresponds to single-image testing (20 images mixed from Mendeley, Roboflow, Google, and validation data), followed by batch evaluation on 200 images (100 TB-positive, 100 TB-negative from Mendeley). The performance is summarized through confusion matrix, and ROC curve. The set of 200 images were selected from the Mendeley Tuberculosis Chest X-rays dataset [18] (Fig. 3).

4. Discussion

The experiments show that even a relatively simple, scratch-trained CNN can deliver a 92% accuracy rate. Importantly, this was achieved on consumer-grade hardware, reinforcing the idea that useful TB screening tools don't always require the massive architectures reported in recent deep learning studies. Comparing it to an earlier study by Showkatian et al. [28], which reported an accuracy of 87% with a scratch-trained CNN, the present model shows improvement in both accuracy and generalization, despite being architecturally simpler. Goswami et al. [29] reported a comparable accuracy of 94% using a scratch-trained CNN trained primarily on Kaggle-sourced radiographs. While the reported accuracy marginally surpasses the current results, it is noteworthy that their dataset was comparatively homogeneous. This lack of diversity potentially limits its applicability across variable real-world clinical imaging conditions. By contrast, the dataset used in this study was formed by merging multiple publicly available repositories, therefore incorporating heterogeneity in acquisition standards, imaging quality, and annotation styles (see Table 1). This deliberate design choice strengthened the model's ability to generalize across unseen samples, an essential quality for clinical deployment [11]. Beyond the training phase, generalization was empirically assessed through a staged testing approach. First, an **initial pilot assessment** was conducted on a small set of 20 radiographs sourced from Mendeley, Roboflow [17, 18], Google Images, and the internal validation split. This allowed for a fine-grained check of prediction confidence at the individual image level. **While these pilot results were promising, they represent a preliminary feasibility check.** To address the limitations of small sample sizes and

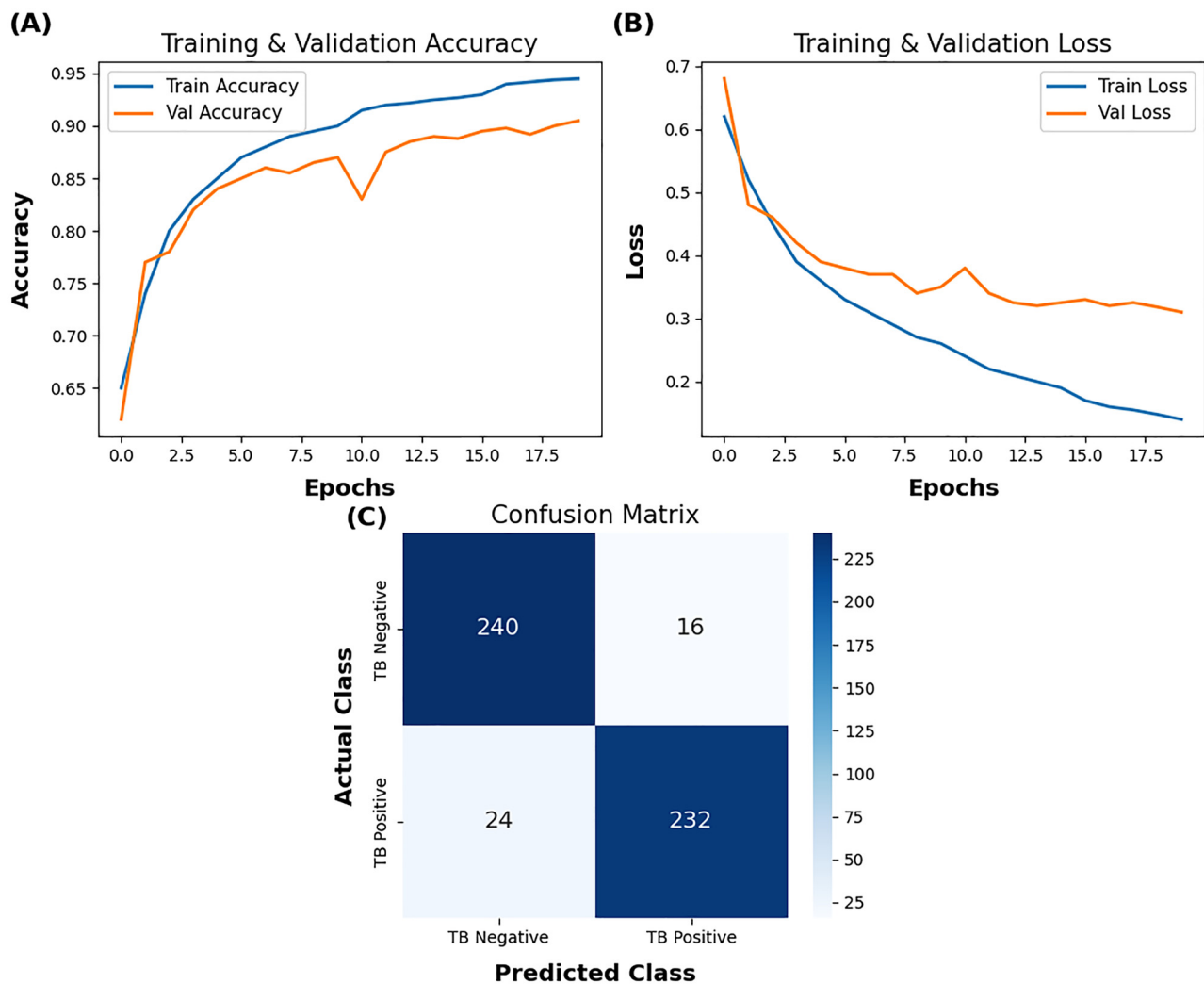


Fig. 2. Model training performance and evaluation on the internal test set. (A) Training and validation accuracy curves plotted across 20 training epochs. (B) Training and validation loss curves, demonstrating model convergence and stability. (C) Confusion matrix detailing the classification performance on the held-out internal test set, showing the counts for correctly and incorrectly classified Normal and TB cases

provide definitive proof of generalization, the system was subsequently evaluated on a **substantially larger independent test set of 200 radiographs** derived from the Mendeley dataset. This larger-scale validation confirmed balanced sensitivity under bulk inference conditions. Advanced models such as TB-Net [7] and ResfEANet [30] achieve significantly higher accuracies of 99.9 and 97.6%, respectively, with the integration of attention mechanisms and automated architectural design. However, these gains are attained at the expense of substantial computational overhead, requiring GPU acceleration and specialized infrastructure. The proposed model, by contrast, achieves competitive performance on consumer-grade hardware (Intel i5 CPU, 16 GB RAM, mid-range GPU), underscoring its potential utility in resource-constrained environments where high-end computational resources are unavailable. Importantly, the integration of Swish activation further enhanced gradient flow and convergence stability, aligning with prior evidence of its effectiveness in biomedical imaging contexts [25].

Moreover, the model demonstrated resilience when tested across radiographs in multiple formats (JPEG, JPG, PNG), highlighting adaptability to diverse clinical workflows where input standards may vary. Subsequently, the system's batch processing capability was evaluated on a larger test set comprising 200 radiographs derived from the Mendeley dataset. The confusion matrix, and ROC curve analysis demonstrated that the model preserved high classification accuracy but also maintained balanced sensitivity and specificity under bulk inference conditions. The results also underscore the importance of computational efficiency: average inference times on the CPU ranged between 30 and 40 ms, confirming feasibility for near-real-time decision support. This complements with the findings of Muthi'ah et al. [31], who showed that CNNs trained for minimal epochs (as few as 10) maintained competitive accuracy while dramatically reducing training time and resource demands, underscoring the value of efficiency-driven design choices for TB screening.

Taken together, these findings indicate that the proposed CNN positions itself as a practical alternative to more

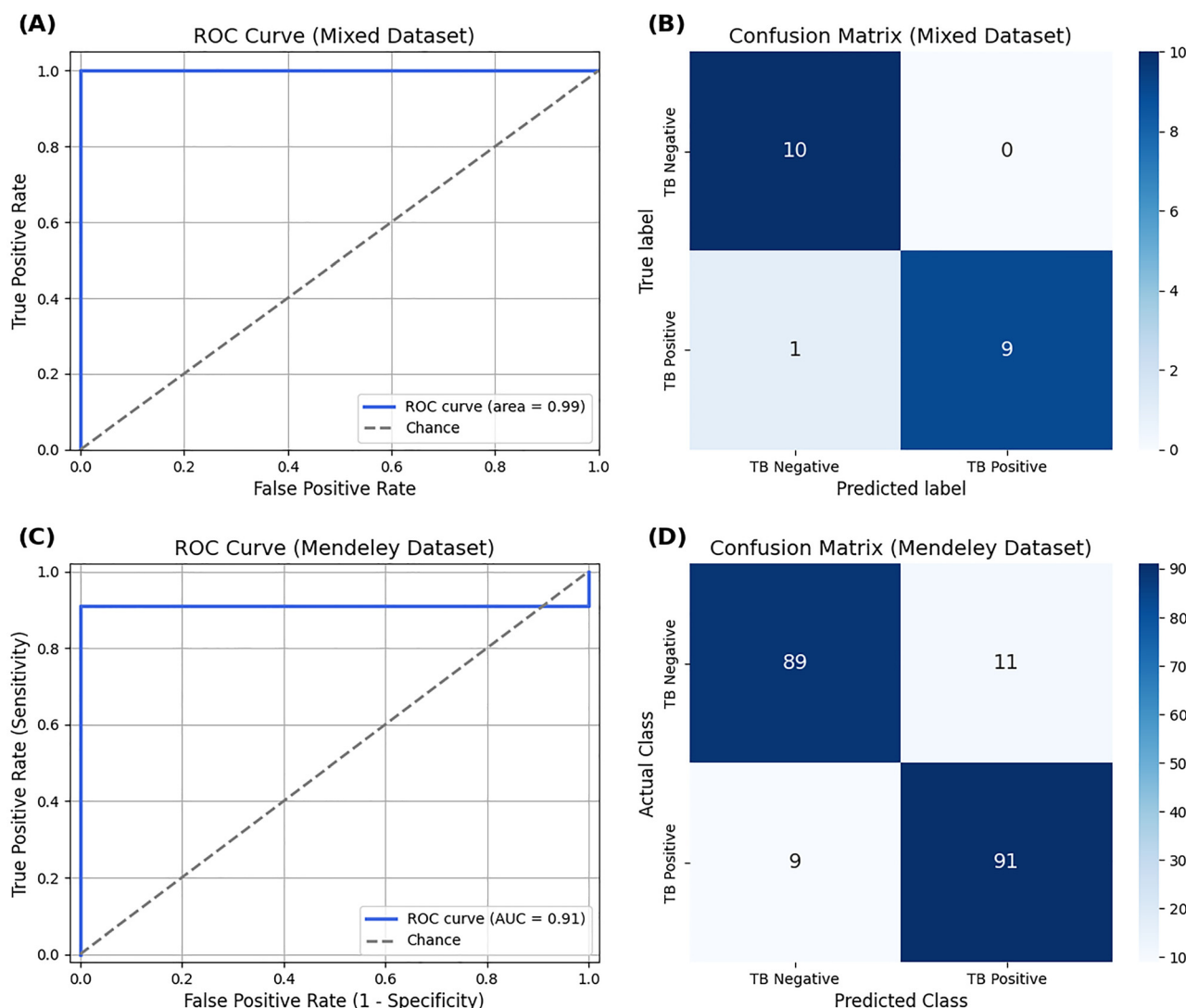


Fig. 3. External validation performance on two independent, unseen datasets. (A) Receiver Operating Characteristic (ROC) curve and (B) confusion matrix for the initial validation on a mixed set of 20 images, achieving an Area Under the Curve (AUC) of 0.99. (C) ROC curve and (D) confusion matrix for the large-scale batch validation on 200 images from the Mendeley dataset, achieving an AUC of 0.91

complex attention-driven architecture. The consistency of performance across both small-scale and large-scale evaluations reinforces the reliability of the proposed CNN. Furthermore, the ROC curve and confusion matrix helped to highlight minimal trade-offs between precision and recall. While it does not match their absolute performance in accuracy, it offers reproducibility, adaptability, and deployment readiness without prohibitive computational demands. This balance between performance and accessibility makes it especially relevant for tuberculosis screening initiatives in low- and middle-income countries, where infrastructure limitations often preclude the use of cutting-edge but resource-intensive systems. The study however has several limitations, including reliance on publicly available datasets with potential labeling noise and the use of a relatively simple model without extensive tuning or augmentation. Its focus on binary classification also limits clinical scope and indicates the need for future work on richer annotations, multi-class tasks, explainable AI.

5. Conclusion

This study presented a custom Convolutional Neural Network (CNN) trained from scratch for automated tuberculosis detection in chest radiographs. The proposed model achieved 92% accuracy, with precision and recall values above 0.90, while remaining computationally efficient and independent of transfer learning. By aggregating a heterogeneous dataset from multiple public repositories, the model was exposed to diverse imaging qualities and annotation standards, strengthening its ability to generalize to real-world settings. Compared with existing scratch-trained CNN, such as the 94% accuracy reported by Goswami et al. [29] on a more homogeneous dataset and the 87% accuracy obtained by Showkatian et al. [28], the present method delivers competitive performance within a lightweight design. Although advanced architectures like ResFEANet [30] and

TB-Net [7] achieve higher raw accuracy through attention mechanisms and neural architecture search, their computational demands limit their applicability in low-resource environments. The results demonstrate that carefully designed, scratch-trained CNNs can offer a practical balance between accuracy, efficiency, and accessibility for TB screening programs, particularly in under-resourced healthcare systems. In summary, the findings indicate that lightweight scratch-trained CNNs hold strong potential for contributing to TB detection. Although they may not match the absolute accuracy of larger architectures, their efficiency and ability to operate on modest computing systems make them highly practical for deployment in clinics and screening programs.

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Ethical approval: Not applicable, as the study used publicly available, anonymized datasets.

AI assistance disclosure: The authors acknowledge the use of AI-based language tools (ChatGPT, OpenAI) solely for grammar refinement and sentence polishing. All research design, experiments, data analysis, and conclusions are the authors' own.

Data availability: All datasets used in this study are publicly available and have been appropriately cited within the manuscript.

Author's contributions: Muhammad Fawad conceptualized and designed the study. Asadullah Baig assisted in data preprocessing and model implementation. Irfan Usmani supervised the project.

All authors reviewed the final version of the manuscript and agreed to submit it to IMAGING for publication.

Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1556/1647.2026.00376>.

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